Table 2 Biopsy information and pathological diagnoses

Characteristic, n (%)	Vitamin C	Control	P value
	N = 355	N = 355	
Needle insertions			
2	350 (98.6)	345 (97.2)	0.192
3	5 (1.4)	10 (2.8)	
Left renal biopsy	0	2 (0.6)	0.499
Primary Renal Diagnosis[a]			
IgA nephropathy	111 (31.3)	103 (31.3)	0.513
Membranous nephropathy	69 (19.4)	86 (24.2)	0.122
Diabetic nephropathy	47 (13.2)	38 (10.7)	0.298
Hypertensive renal disease[b]	34 (9.6)	21 (5.9)	0.068
Minimal change disease	15 (4.2)	17 (4.8)	0.718
Focal segmental glomerulosclerosis	12 (3.4)	15 (4.2)	0.556
Lupus nephritis	6 (1.7)	8 (2.3)	0.589
Other glomerulonephritides[c]	47 (13.2)	51 (14.4)	0.663

[a]In cases of multiple diagnostic entries, only the highest-priority diagnosis was selected for classification

[b]Hypertensive renal disease was classified as such only when it was the primary or sole diagnosis

[c]Includes all remaining diagnostic categories with individually low frequencies. Full distribution provided in Supplementary Table 2

Table 3 Primary Outcomes

Analysis model	Vitamin C group	Control group	Effect size (95% CI)	P value
	N = 355	N = 355		
Perirenal hematoma				
All Participants				
Any hematoma, n (%)	200 (56.3)	178 (50.1)	RR:1.124 (0.978, 1.290)	0.098
Crude area, mm^2	191.02 ± 274.89	189.37 ± 466.36	MD:1.645 (−54.764, 58.054)	0.954
Adjusted mean area[a], mm^2	192.42	187.97	MD:4.435 (−52.305, 61.176)	0.878
Participants with Hematoma				
Area, mm^2	339.06 ± 289.78	377.68 ± 602.87	MD:−38.625 (−132.690, 55.440)	0.420
Pain VAS scores				
2 h[b]	0 [0, 9]	0 [0, 6]	–	0.105
12 h[b]	0 [0, 7]	0 [0, 5]	–	0.192
24 h[b]	0 [0, 4]	0 [0, 3]	–	0.534

[a]Adjusted for baseline age by ANCOVA. Values presented are estimated marginal means. Effect size is mean difference (MD) or risk ratio (RR)

[b]Data are presented as median [IQR] and were compared using the Mann-Whitney U test. Adjusted P-values were calculated using the Bonferroni correction for comparisons across 3 time points

participants (74.6% vs. 67.3%). At 2 h, a non-significant trend was observed (P = 0.095). By 24 h, the distribution between groups was similar, with no evidence of a linear trend (P = 0.459).

Changes in laboratory and vital sign parameters

Postbiopsy urinary vitamin C concentration increased significantly in the vitamin C group compared with the control group (Diff, 25.67 ± 18.09 mg/L vs. −1.01 ± 9.66 mg/L; SMD, 1.839; 95% CI, 1.663 to 2.014; P < 0.001), confirming adherence to the protocol. No significant between-group differences were observed in the postbiopsy changes from baseline in systolic blood pressure, hemoglobin level, or platelet count (Table 5).

Subgroup analyses

In post-hoc exploratory subgroup analyses (Additional file 1: Table S7–S9), we observed no significant interaction between treatment and baseline urinary vitamin C levels for the primary outcome of hematoma area (tertiles p = 0.422; quartiles p = 0.484) or for pain outcomes (all p for interaction >0.05). No subgroup benefited from vitamin C.

Adverse events

Adverse events were monitored from the first administration of the intervention until the end of study. The overall incidence of AEs was low and comparable between the two groups: 4.2% (15 of 355) in the Vitamin C group and 3.7% (13 of 355) in the control group (Table 6 and Additional file 1: Table S5). Gastrointestinal disorders were the most frequently reported events. All reported AEs